

Notes on Recursive Utility

François Gourio*

References: Epstein-Zin (Ecta 1989, JPE 1991), Kreps-Porteus (Ecta 1978), Tallarini (JME 2000), Hansen-Sargent-Tallarini REStud 1999, Hansen-Sargent 1995, Hansen-Heaton-Li 2005, Weil REStud 1993, QJE 1989, JME 1989, Campbell AER 1993, JPE 1996, Bansal and Yaron JF 2004.

1 Basics

EZ following work by Kreps and Porteus introduced a class of preferences which allow to break the link between risk aversion and intertemporal substitution. Preferences are defined recursively over current (known) consumption and a certainty equivalent of tomorrow's utility:

$$V_t = F(c_t, R_t(V_{t+1})). \quad (1.1)$$

$$R_t(V_{t+1}) = G^{-1}(E_t G(V_{t+1})), \quad (1.2)$$

with F and G increasing and concave.

Most of the literature considers simple functional forms for F and G :

$$\begin{aligned} \rho > 0 : F(c, z) &= ((1 - \beta)c^{1-\rho} + \beta z^{1-\rho})^{\frac{1}{1-\rho}}, \\ \alpha > 0 : G(x) &= \frac{x^{1-\alpha}}{1-\alpha}. \end{aligned}$$

*Boston University, Department of Economics. Email: fgourio@bu.edu.

Note I will use the following limits:¹

$$\begin{aligned}\rho &= 1 : F(c, z) = c^{1-\beta} z^\beta. \\ \alpha &= 1 : G(x) = \log x.\end{aligned}$$

Hence

$$\begin{aligned}\alpha > 0 : R_t(V_{t+1}) &= E_t(V_{t+1}^{1-\alpha})^{\frac{1}{1-\alpha}}, \\ \alpha = 1 : R_t(V_{t+1}) &= \exp(E_t \log(V_{t+1})).\end{aligned}$$

Special case: if $\alpha = \rho$ or if consumption is deterministic: we have the usual standard time-separable expected discounted utility with discount factor β and IES = $\frac{1}{\rho}$, risk aversion $\alpha = \rho$.

- In general α is the relative risk aversion coefficient for static gambles and ρ is the inverse of the intertemporal elasticity of substitution for deterministic variations.
- Discuss simple example with two lotteries:

-lottery A pays in each period $t = 1, 2, \dots$ c_h or c_l , the probability is $\frac{1}{2}$ and the outcome is iid across period;

- lottery B pays starting at $t = 1$ either c_h at all future dates for sure, or c_l at all future date for sure; there is a single draw at time $t = 1$.

“Intertemporal composition of risk matters.”: we cannot reduce compound lotteries.

- Early resolution of uncertainty is preferred if and only if $\alpha > \rho$.

Result: The stochastic discount factor is

$$m_{t,t+1} = \beta \left(\frac{c_{t+1}}{c_t} \right)^{-\rho} \left(\frac{V_{t+1}}{R_t(V_{t+1})} \right)^{\rho-\alpha}.$$

Proof:

1

$$\begin{aligned}((1-\beta)c^{1-\rho} + \beta z^{1-\rho})^{\frac{1}{1-\rho}} &= \exp\left(\frac{1}{1-\rho} \log\left(1 + (1-\rho)\left((1-\beta)\frac{c^{1-\rho}-1}{1-\rho} + \beta\frac{z^{1-\rho}-1}{1-\rho}\right)\right)\right) \\ &\simeq \exp\left((1-\beta)\frac{c^{1-\rho}-1}{1-\rho} + \beta\frac{z^{1-\rho}-1}{1-\rho}\right) \\ &\simeq \exp((1-\beta)\ln c + \beta\ln z) \\ &\simeq c^{1-\beta} z^\beta.\end{aligned}$$

Start with the definition of m as the IMRS:

$$m_{t,t+1}(s^t, s_{t+1}) = \frac{\partial V_t / \partial c_{t+1}(s^t, s_{t+1})}{\partial V_t / \partial c_t(s^t)}.$$

Differentiation of (1.1) w.r.t. c_t :

$$\frac{\partial V_t}{\partial c_t} = F_1(c_t, R_t(V_{t+1})).$$

Differentiation of (1.1) w.r.t. $c_{t+1}(s^t, s_{t+1})$:

$$\begin{aligned} \frac{\partial V_t}{\partial c_{t+1}(s^t, s_{t+1})} &= F_2(c_t, R_t(V_{t+1})) \times \frac{\partial R_t(V_{t+1})}{\partial V_{t+1}(s^t, s_{t+1})} \times \frac{\partial V_{t+1}(s^t, s_{t+1})}{\partial c_{t+1}(s^t, s_{t+1})} \\ &= F_2(c_t, R_t(V_{t+1})) \times (E_t V_{t+1}^{1-\alpha})^{\frac{1}{1-\alpha}-1} \times V_{t+1}^{-\alpha} \times F_1(c_{t+1}(s^t, s_{t+1}), R_{t+1}(V_{t+2})). \end{aligned}$$

since

$$\frac{\partial R_t(V_{t+1})}{\partial V_{t+1}(s^t, s_{t+1})} = (E_t V_{t+1}^{1-\alpha})^{\frac{1}{1-\alpha}-1} V_{t+1}^{-\alpha}.$$

Note now:

$$\begin{aligned} F_1(c_t, R_t(V_{t+1})) &= (1-\beta)c_t^{-\rho} ((1-\beta)c_t^{1-\rho} + \beta R_t(V_{t+1})^{1-\rho})^{\frac{1}{1-\rho}-1} \\ &= (1-\beta)c_t^{-\rho} F(c_t, R_t(V_{t+1}))^\rho. \end{aligned}$$

$$\begin{aligned} F_2(c_t, R_t(V_{t+1})) &= \beta R_t(V_{t+1})^{-\rho} F(c_t, R_t(V_{t+1}))^\rho, \\ &= \beta R_t(V_{t+1})^{-\rho} V_t^\rho. \end{aligned}$$

I get finally

$$\begin{aligned} m_{t,t+1}(s^t, s_{t+1}) &= \frac{F_2(c_t, R_t(V_{t+1})) \times (E_t V_{t+1}^{1-\alpha})^{\frac{1}{1-\alpha}-1} \times V_{t+1}^{-\alpha} \times F_1(c_{t+1}(s^t, s_{t+1}), R_{t+1}(V_{t+2}))}{F_1(c_t, R_t(V_{t+1}))} \\ &= \frac{\beta R_t(V_{t+1})^{-\rho} F(c_t, R_t(V_{t+1}))^\rho (E_t V_{t+1}^{1-\alpha})^{\frac{1}{1-\alpha}-1} V_{t+1}^{-\alpha} c_{t+1}^{-\rho} F(c_{t+1}, R_{t+1}(V_{t+2}))^\rho}{c_t^{-\rho} F(c_t, R_t(V_{t+1}))^\rho} \\ &= \beta \left(\frac{c_{t+1}}{c_t} \right)^{-\rho} \left[R_t(V_{t+1})^{-\rho} \times (E_t V_{t+1}^{1-\alpha})^{\frac{1}{1-\alpha}-1} \times V_{t+1}^{-\alpha} F(c_{t+1}, R_{t+1}(V_{t+2}))^\rho \right] \\ &= \beta \left(\frac{c_{t+1}}{c_t} \right)^{-\rho} \left[R_t(V_{t+1})^{-\rho} \times R_t(V_{t+1})^\alpha \times V_{t+1}^{-\alpha} \times V_{t+1}^\rho \right] \\ &= \beta \left(\frac{c_{t+1}}{c_t} \right)^{-\rho} \left(\frac{V_{t+1}}{R_t(V_{t+1})} \right)^{\rho-\alpha} \square \end{aligned}$$

2 Empirical implementation (Epstein-Zin 1991)

Our formula for the SDF has an obvious problem: it depends on future utilities $\frac{V_{t+1}}{R_t(V_{t+1})}$, which are not observed. The first possible empirical implementation of these preferences uses the fact that the *return on wealth* can be substituted in instead of future utility (the second term of the discount factor). This allows to do (i) GMM estimation as Hansen-Singleton (1983), or (ii) to use log-linear-log-normal approximations.

The derivation is as follows. *Define* wealth as the PDV of consumption, or recursively as:

$$W_t = c_t + E_t(m_{t,t+1}W_{t+1}).$$

Result: $W_t = V_t / F_1(c_t, R_t(V_{t+1}))$.

Proof: Guess and verify that satisfies this recursion.

$$\begin{aligned} W_t &= c_t + E_t(m_{t,t+1}W_{t+1}) \\ &= c_t + E_t\left(\beta\left(\frac{c_{t+1}}{c_t}\right)^{-\rho}\left(\frac{V_{t+1}}{R_t(V_{t+1})}\right)^{\rho-\alpha}\frac{V_{t+1}}{(1-\beta)c_{t+1}^{-\rho}F(c_{t+1}, R_{t+1}(V_{t+2}))^\rho}\right) \\ &= c_t + E_t\left(\beta\left(\frac{c_{t+1}}{c_t}\right)^{-\rho}V_{t+1}^{1-\alpha}R_t(V_{t+1})^{\alpha-\rho}\frac{1}{(1-\beta)c_{t+1}^{-\rho}}\right) \\ &\quad \frac{V_t}{F_1(c_t, R_t(V_{t+1}))}(1-\beta)c_t^{-\rho}R_t(V_{t+1})^{\rho-\alpha} \stackrel{?}{=} (1-\beta)c_t^{1-\rho}R_t(V_{t+1})^{\rho-\alpha} + \beta E_t(V_{t+1}^{1-\alpha}) \\ &\quad \frac{V_t}{(1-\beta)c_t^{-\rho}F(c_t, R_t(V_{t+1}))^\rho}(1-\beta)c_t^{-\rho}R_t(V_{t+1})^{\rho-\alpha} \stackrel{?}{=} (1-\beta)c_t^{1-\rho}R_t(V_{t+1})^{\rho-\alpha} + \beta E_t(V_{t+1}^{1-\alpha}) \\ &\quad V_t^{1-\rho}R_t(V_{t+1})^{\rho-\alpha} \stackrel{?}{=} (1-\beta)c_t^{1-\rho}R_t(V_{t+1})^{\rho-\alpha} + \beta E_t(V_{t+1}^{1-\alpha}) \\ &\quad V_t^{1-\rho} \stackrel{?}{=} (1-\beta)c_t^{1-\rho} + \beta R_t(V_{t+1})^{1-\rho}, \end{aligned}$$

which is true. This confirms our guess. $\square$

Note that I can now define the return on the wealth portfolio:

$$R_{t,t+1} = \frac{W_{t+1}}{W_t - C_t}.$$

Simple algebra shows that $R_{t,t+1} = \left\{\beta\left(\frac{c_{t+1}}{c_t}\right)^{-\rho}\left(\frac{R_t(V_{t+1})}{V_{t+1}}\right)^{1-\rho}\right\}^{-1}$.

Proof:

$$\begin{aligned}
R_{t,t+1} &= \frac{V_{t+1}}{F_1(c_{t+1}, R_{t+1}(V_{t+2}))} \frac{1}{\left(\frac{V_t}{F_1(c_t, R_t(V_{t+1}))} - C_t\right)} \\
&= \frac{V_{t+1}}{F_1(c_{t+1}, R_{t+1}(V_{t+2}))} \frac{F_1(c_t, R_t(V_{t+1}))}{(V_t - C_t F_1(c_t, R_t(V_{t+1})))} \\
&= \frac{V_{t+1}}{(1-\beta)c_{t+1}^{-\rho} V_{t+1}^\rho} \frac{(1-\beta)c_t^{-\rho} F(c_t, R_t(V_{t+1}))^\rho}{\beta R_t(V_{t+1})^{1-\rho} F(c_t, R_t(V_{t+1}))^\rho} \\
R_{t,t+1} &= \left\{ \beta \left(\frac{c_{t+1}}{c_t}\right)^{-\rho} \left(\frac{R_t(V_{t+1})}{V_{t+1}}\right)^{1-\rho} \right\}^{-1}. \square
\end{aligned}$$

Hence the SDF can be expressed as a function of the return:

$$\begin{aligned}
m_{t,t+1} &= \beta \left(\frac{c_{t+1}}{c_t}\right)^{-\rho} \left(\frac{V_{t+1}}{R_t(V_{t+1})}\right)^{\rho-\alpha} \\
&= \beta \left(\frac{c_{t+1}}{c_t}\right)^{-\rho} \left\{ \frac{R_{t,t+1}^{-1}}{\beta \left(\frac{c_{t+1}}{c_t}\right)^{-\rho}} \right\}^{\frac{\rho-\alpha}{\rho-1}} \\
&= \beta \left(\frac{c_{t+1}}{c_t}\right)^{-\rho \left(1-\frac{\rho-\alpha}{\rho-1}\right)} R_{t,t+1}^{\frac{\rho-\alpha}{1-\rho}} \\
&= \beta \left(\frac{c_{t+1}}{c_t}\right)^{-\rho \frac{\alpha-1}{\rho-1}} R_{t,t+1}^{\frac{\rho-\alpha}{1-\rho}}.
\end{aligned}$$

This allows to implement this empirically since the SDF can now be measured. Epstein and Zin proxy the return on wealth by the return on a broad stock market return. An obvious criticism is that the a lot of wealth is not traded on the stock market (private firms, human capital, housing...) but these returns may be correlated with the stock market return. Future work (eg Campbell 1996 JPE) try to add human capital, and we will talk about this in the next class.

Note that assets' risk are measured as the covariance with the SDF, so the Epstein-Zin utility rationalizes a formula which is a mix of the CAPM and the CCAPM:

$$\log \left(\frac{ER^i}{R^f} \right) = \rho Cov(R^i, \Delta \log C) + \frac{\rho - \alpha}{1 - \alpha} Cov(R^i, R^m).$$

Empirically, the extra free parameter of EZ allows to improve on CRRA. However the solutions of asset pricing puzzles with EZ utility require high risk aversion (except, to some extent, the Bansal-Yaron paper we will mention later).

A remark on the consumption-wealth ratio: given that

$$\begin{aligned}
W_t &= \frac{V_t}{F_1(c_t, R_t(V_{t+1}))} \\
&= \frac{V_t}{(1-\beta)c_t^{-\rho} F(c_t, R_t(V_{t+1}))^\rho} \\
&= \frac{V_t^{1-\rho}}{(1-\beta)c_t^{-\rho}}. \\
c_t &= (1-\beta)W_t \frac{c_t^{1-\rho}}{V_t^{1-\rho}},
\end{aligned}$$

so that if $\rho = 1$ (log utility in IES), the consumption-wealth ratio is constant. Empirical work in Lettau and Ludvigson (2001 JPE) gives an important role to this variable.

3 Tallarini/Hansen computations when IES=1

Another approach to implement Epstein-Zin utility empirically is to solve exactly for the value V_t , given a tractable consumption process. This implies we do not need to introduce the market return in the SDF. This section does this computation in the special case $\rho = 1$, i.e. an *IES* equal to unity, and for arbitrary autocorrelations of consumption growth. There is a fair bit of algebra but I give all of this to you for free. First, write the recursion for $\sigma = 1$:

$$V_t = C_t^{1-\beta} (E_t V_{t+1}^{1-\alpha})^{\frac{\beta}{1-\alpha}}$$

$$\begin{aligned}
\log V_t &= (1-\beta) \log C_t + \frac{\beta}{1-\alpha} \log E_t (V_{t+1}^{1-\alpha}) \\
\log V_t &= (1-\beta) \log C_t + \frac{\beta}{1-\alpha} \log E_t \exp((1-\alpha) \log V_{t+1})
\end{aligned}$$

Assume

$$\begin{aligned}
\Delta \log C_t &= \gamma(L)w_t + \mu_c \\
w_t &\text{ iid } N(0, I)
\end{aligned}$$

From our previous computations, the stochastic discount factor is:

$$m_{t,t+1} = \beta \frac{c_t}{c_{t+1}} \frac{\exp((1-\alpha) \ln V_{t+1})}{E_t \exp((1-\alpha) \ln V_{t+1})}$$

We now guess and verify that $v_t = \xi c_t + \alpha(L)w_t + \mu_v$ (where lowercase denotes log). Verification:

$$\begin{aligned}
& E_t \exp((1 - \alpha)v_{t+1}) \\
&= E_t \exp((1 - \alpha)(\xi c_{t+1} - \xi c_t + \xi c_t + \alpha(L)w_{t+1} + \mu_v)) \\
&= \exp((1 - \alpha)(\xi c_t + \mu_v + \xi \mu_c)) E_t \exp\left(\frac{\sigma}{2}((\xi \gamma(L) + \alpha(L))w_{t+1})\right) \\
&= \exp((1 - \alpha)(\xi c_t + \mu_v + \xi \mu_c)) \exp\left((1 - \alpha)E_t(\xi \gamma(L) + \alpha(L))w_{t+1} + \frac{(1 - \alpha)^2}{2}(\xi \gamma(0) + \alpha(0))'(\xi \gamma(0) + \alpha(0))\right) \\
&= \exp((1 - \alpha)(\xi c_t + \mu_v + \xi \mu_c)) \exp\left((1 - \alpha)\left[\frac{\xi \gamma(L) + \alpha(L)}{L}\right]_+ w_t + \frac{(1 - \alpha)^2}{2}(\xi \gamma(0) + \alpha(0))'(\xi \gamma(0) + \alpha(0))\right)
\end{aligned}$$

$$\begin{aligned}
v_t &= (1 - \beta)c_t + \frac{\beta}{1 - \alpha} \log E_t \exp((1 - \alpha)v_{t+1}) \\
&= (1 - \beta)c_t + \beta(\xi c_t + \mu_v + \xi \mu_c) + \beta \left[\frac{\xi \gamma(L) + \alpha(L)}{L}\right]_+ w_t + \frac{\beta(1 - \alpha)}{2}(\xi \gamma(0) + \alpha(0))'(\xi \gamma(0) + \alpha(0))
\end{aligned}$$

$$v_t = (1 - \beta + \beta\xi)c_t + \beta(\mu_v + \xi \mu_c) + \beta \left[\frac{\xi \gamma(L) + \alpha(L)}{L}\right]_+ w_t + \frac{\beta(1 - \alpha)}{2}(\xi \gamma(0) + \alpha(0))'(\xi \gamma(0) + \alpha(0))$$

Where $[A(L)]_+$ = terms with nonnegative degrees of this polynomial fraction. (See the section on “Hansen-Sargent prediction formulas” for some explanations.)

Identification with $v_t = \xi c_t + \alpha(L)w_t + \mu_v$:

$$\begin{aligned}
\xi &= (1 - \beta + \beta\xi) \Rightarrow \xi = 1, \\
\mu_v &= \beta(\mu_v + \xi \mu_c) + \frac{\beta(1 - \alpha)}{2}(\xi \gamma(0) + \alpha(0))'(\xi \gamma(0) + \alpha(0)) \\
&\Rightarrow \mu_v(1 - \beta) = \beta\mu_c + \frac{\beta\sigma}{4}(\gamma(0) + \alpha(0))'(\gamma(0) + \alpha(0)),
\end{aligned}$$

and

$$\alpha(L) = \beta \left[\frac{\gamma(L) + \alpha(L)}{L}\right]_+$$

Clearly ξ is uniquely determined and μ_v is also given $\alpha(L)$. To see how the last equation fully determines $\alpha(L)$, it is useful to use z -transforms:

$$\alpha(z) = \beta \frac{\gamma(z) + \alpha(z) - \alpha(0) - \gamma(0)}{z}$$

$$\alpha(\beta) = \gamma(\beta) + \alpha(\beta) - \alpha(0) - \gamma(0)$$

$$\alpha(0) = \gamma(\beta) - \gamma(0)$$

$$\begin{aligned}
\alpha(z)z &= \beta(\alpha(z) + \gamma(z) - \gamma(\beta)) \\
\alpha(z)(z - \beta) &= \gamma(z) - \gamma(\beta) \\
\alpha(z) &= \frac{\gamma(z) - \gamma(\beta)}{z - \beta},
\end{aligned}$$

which has a pole for $z = \beta$, so this is a polynomial with only nonnegative degrees. $\square$

Computing asset prices

Given the expression for m , it is not too hard to compute the risk free rate and the risk premium. The key result is the following. Since dividends and consumption are jointly log-normal, the risk premium is given by:

$$\begin{aligned}
\log E_t \frac{R_{i,t+1}}{R_{f,t+1}} &= E_t \log \frac{R_{i,t+1}}{R_{f,t+1}} + \frac{1}{2} \text{Var}_t \log R_{i,t+1} \\
&= -\text{Cov}_t(\log R_{i,t+1}, \log m_{t,t+1}) \\
&= -\text{Cov}_t(\log R_{i,t+1}, \log m_{t,t+1} - E_t \log m_{t,t+1})
\end{aligned}$$

In our case,

$$\begin{aligned}
\log m_{t+1} - E_t \log m_{t+1} &= -(\Delta \log C_{t+1} - E_t \Delta \log C_{t+1}) + (1 - \alpha)(v_{t+1} - E_t v_{t+1}) \\
&= -\gamma(0)w_{t+1} + (1 - \alpha)(\gamma(0)w_{t+1} + \alpha(0)w_{t+1}) \\
&= [-\gamma(0) + (1 - \alpha)\gamma(\beta)]w_{t+1}.
\end{aligned}$$

Where the last line follows from $\alpha(0) = \gamma(\beta) - \gamma(0)$. (Note my poor notation: I have two alpha: risk aversion and the lag polynomial $\alpha(L)$.) The first term $\gamma(0)$ is the standard one (risk premium associated with log utility). The second one is new and depends on the future path of consumption as measured by $\gamma(\beta) = \sum_{j \geq 0} \gamma_j \beta^j$. This is the *long-run risk* which now plays a role with EZ utility. It is directly related to the idea that the ‘intertemporal composition of risk matters’. Note that if $\beta \rightarrow 1$, this term can be substantial and high for high risk aversion.

4 Log-linearization (Campbell)

This is the third main approach to Epstein-Zin utility. Campbell uses some log-linear approximations to derive implications even the IES is not one. (Some of these log-linear approximations can be useful also with CRRA, or for empirical work.)

Just like Campbell and Shiller did a log-linear approximation of the return, Campbell, writes a log-linear approximation to the budget constraint. He writes the budget constraint as:

$$W_{t+1} = R_{t+1}^m (W_t - C_t)$$

Now do all the steps similar to Campbell-Shiller:

$$\begin{aligned}\frac{W_{t+1}}{W_t} &= R_{t+1}^m \left(1 - \frac{C_t}{W_t}\right) \\ \Delta \log W_{t+1} = w_{t+1} - w_t &= r_{t+1}^m + \log \left(1 - \frac{C_t}{W_t}\right) \\ \log \left(1 - \frac{C_t}{W_t}\right) &= \log (1 - \exp (c_t - w_t)) \\ &\simeq k + \left(1 - \frac{1}{\rho}\right) (c_t - w_t). \\ \rho &= \frac{W - C}{W} < 1\end{aligned}$$

This yields:

$$\Delta w_{t+1} = r_{t+1}^m + k + \left(1 - \frac{1}{\rho}\right) (c_t - w_t) \quad (4.1)$$

We can rewrite this equation as:

$$\begin{aligned}\Delta c_{t+1} + (c_t - w_t) - (c_{t+1} - w_{t+1}) &= r_{t+1}^m + k + \left(1 - \frac{1}{\rho}\right) (c_t - w_t) \\ \frac{1}{\rho}(c_t - w_t) - (c_{t+1} - w_{t+1}) &= r_{t+1}^m + k - \Delta c_{t+1} \\ (c_t - w_t) - \rho(c_{t+1} - w_{t+1}) &= \rho(r_{t+1}^m + k - \Delta c_{t+1})\end{aligned} \quad (4.2)$$

Iterating forward yields:

$$\begin{aligned}c_t - w_t &= \sum_{j \geq 1} \rho^j (r_{t+j}^m + k - \Delta c_{t+j}), \\ &= \frac{k\rho}{1 - \rho} + \sum_{j \geq 1} \rho^j (r_{t+j}^m - \Delta c_{t+j}).\end{aligned} \quad (4.3)$$

This is just an *accounting identity*, which holds ex-post as well as ex-ante. This holds also in expectation:

$$c_t - w_t = \frac{k\rho}{1 - \rho} + E_t \sum_{j \geq 1} \rho^j (r_{t+j}^m - \Delta c_{t+j}).$$

When the consumption-wealth ratio is high, it means that either future returns will be high or future consumption growth will be low (so that the C/W ratio returns to its average).

Another way to state this equality is to apply the operator $E_{t+1} - E_t$ to equation 4.2 (this operator cancels all terms known at time t):

$$(E_{t+1} - E_t) \Delta c_{t+1} = (E_{t+1} - E_t) (c_{t+1} - w_{t+1}) + (E_{t+1} - E_t) r_{t+1}^m$$

$$\begin{aligned}
c_{t+1} - E_t c_{t+1} &= (E_{t+1} - E_t) \sum_{j \geq 1} \rho^j (r_{t+1+j}^m - \Delta c_{t+j}) + r_{t+1}^m - E_t r_{t+1}^m \\
&= (E_{t+1} - E_t) \sum_{j \geq 0} \rho^j r_{t+1+j}^m - (E_{t+1} - E_t) \sum_{j \geq 1} \rho^j \Delta c_{t+1+j}.
\end{aligned}$$

Euler equations

Up to now we have only log-linearized the budget constraint. We can use this empirically to measure what explains changes in consumption (just like we did with returns), but the interesting part is to note that consumption, wealth and returns are also tied by the optimality of consumer choice. To do this, we assume that all second moments (variances and covariances) are constant. (Note the consistency problem since we are interested in explaining changing returns...)

Recall the SDF formula with EZ utility when we substitute out the return on wealth:

$$m_{t,t+1} = \beta \left(\frac{c_{t+1}}{c_t} \right)^{-\frac{\rho(\alpha-1)}{\rho-1}} R_{m,t+1}^{\frac{\rho-\alpha}{1-\rho}},$$

When we write the Euler equation for the market return, we obtain:

$$E_t \left(\beta \left(\frac{c_{t+1}}{c_t} \right)^{-\frac{\rho}{\rho-1}(\alpha-1)} R_{m,t+1}^{\frac{\rho-\alpha}{1-\rho}+1} \right) = 1.$$

This leads to:

$$\log \beta - \rho \frac{\alpha-1}{\rho-1} E_t \Delta \log c_{t+1} + \frac{1-\alpha}{1-\rho} E_t \log R_{m,t+1} + \zeta = 0,$$

where ζ is a constant regrouping the (constant) conditional variances and covariances. Hence a first result:

$$E_t \Delta \log c_{t+1} = k + \frac{1}{\rho} E_t r_{m,t+1},$$

where k is a constant. Expected consumption growth moves proportionally to the expected stock return, with the IES $1/\rho$ governing the proportionality.

I now derive the equation for the excess returns. Start from the EE for any return and for the risk-free rate:

$$\begin{aligned}
E_t (m_{t+1} R_{i,t+1}) &= 1 \\
E_t \log m_{t+1} + \frac{1}{2} V_t \log m_{t+1} + E_t \log R_{i,t+1} + \frac{1}{2} V_t \log R_{i,t+1} + Cov_t (\log m_{t+1}, \log R_{i,t+1}) &= 0 \\
E_t \log m_{t+1} + \frac{1}{2} V_t \log m_{t+1} + E_t \log R_{f,t+1} &= 0
\end{aligned}$$

Subtracting these equations yields

$$\log E_t \frac{R_{i,t+1}}{R_{f,t+1}} = E_t \log \frac{R_{i,t+1}}{R_{f,t+1}} + \frac{1}{2} V_t \log R_{i,t+1} \quad (4.4)$$

$$= -Cov_t (\log m_{t+1}, \log R_{i,t+1}) \quad (4.5)$$

$$= \frac{\rho(\alpha-1)}{\rho-1} Cov_t (\Delta \log c_{t+1}, \log R_{i,t+1}) + \frac{\rho-\alpha}{1-\rho} Cov_t (\log R_{m,t+1}, \log R_{i,t+1}) \quad (4.6)$$

You can see how both consumption growth and the market return are risk factors in this equation.

(Campbell uses a different notation, but he has the same results: $\sigma = \frac{1}{\rho}$, $\alpha = \gamma$, and $\theta = \frac{1-\gamma}{1-\frac{\gamma}{\sigma}} = \frac{1-\alpha}{1-\rho}$. So $1 - \theta = \frac{\rho-\alpha}{1-\rho}$ and $\frac{\theta}{\sigma} = \theta\rho = \frac{(1-\alpha)\rho}{1-\rho}$.)

Campbell's last step: use the consumption EE $E_t \Delta c_{t+1} = k + \frac{1}{\rho} E_t r_{m,t+1}$ in the PVBC found above:

$$c_{t+1} - E_t c_{t+1} = (E_{t+1} - E_t) \sum_{j \geq 0} \rho^j r_{t+j}^m - (E_{t+1} - E_t) \sum_{j \geq 1} \rho^j \Delta c_{t+1+j}.$$

Thus

$$\begin{aligned} c_{t+1} - E_t c_{t+1} &= (E_{t+1} - E_t) \sum_{j \geq 0} \rho^j r_{t+1+j}^m - \frac{1}{\rho} (E_{t+1} - E_t) \sum_{j \geq 1} \rho^j \Delta r_{m,t+1+j} \\ &= r_{m,t+1} - E_t r_{m,t+1} + \left(1 - \frac{1}{\rho}\right) (E_{t+1} - E_t) \sum_{j \geq 1} \rho^j r_{m,t+1+j}. \end{aligned} \quad (4.7)$$

We can use this equation in 4.6 to substitute out consumption. Campbell likes this because he believes consumption is a bad proxy for marginal utility and is poorly measured. Hence we get what he calls the 'CAPM+' formula:

$$\begin{aligned} \log E_t \frac{R_{i,t+1}}{R_{f,t+1}} &= \left(\frac{\rho(\alpha - 1)}{\rho - 1} + \frac{\rho - \alpha}{\rho - 1} \right) Cov_t(r_{m,t+1}, r_{i,t+1}) + \frac{\rho(\alpha - 1)}{\rho - 1} \left(1 - \frac{1}{\rho}\right) Cov_t \left((E_{t+1} - E_t) \sum_{j \geq 1} \rho^j r_{m,t+1+j}, r_{i,t+1} \right) \\ &= \alpha Cov_t(r_{m,t+1}, r_{i,t+1}) + (\alpha - 1) Cov_t \left((E_{t+1} - E_t) \sum_{j \geq 1} \rho^j r_{m,t+1+j}, r_{i,t+1} \right) \end{aligned}$$

The novelty is that expectations of future returns now matter. Investors dislike assets that do badly when the market does badly (the first term, which is just the usual CAPM effect), but they also like/dislike assets which do badly when expected future returns are bad. Whether this is a 'like' or 'dislike' depends on whether α is greater or smaller than 1.

This formula can be implemented empirically if you use a VAR to measure the news to future market returns, $(E_{t+1} - E_t) \sum_{j \geq 1} \rho^j r_{m,t+1+j}$.

Note an implication for the BC from the EE: (see Favero WP 2005):

$$\begin{aligned} c_t - w_t &= \frac{k\rho}{1-\rho} + E_t \sum_{j \geq 1} \rho^j (r_{t+j}^m - \Delta c_{t+j}) \\ &= cst + E_t \sum_{j \geq 1} \rho^j \left(r_{m,t+j} - \frac{1}{\rho} r_{m,t+j} \right) \\ &= cst + (1 - \sigma) E_t \sum_{j \geq 1} \rho^j r_{m,t+j} \end{aligned}$$

[nb not the same ρ !] Hence $\sigma < 1$ iff low cw predicts high returns. In the data, Lettau and Ludvigson (2001 JPE) show that a high consumption-wealth ratio forecasts low returns. This suggests we need a low IES.

5 Bansal and Yaron and Long-Run risks

Bansal and Yaron JF 2005 study the implications of recursive utility with $IES > 1$ and risk aversion > 1 , when consumption growth has a highly persistent component, e.g.:

$$\begin{aligned}\Delta \log C_t &= \mu + x_t + u_t, \\ x_t &= \rho x_{t-1} + v_t.\end{aligned}$$

A persistent negative shock to consumption growth (i.e. a negative v) decreases asset prices strongly. (TBA)

6 Disasters

Suppose consumption growth follows the following process:

$$\begin{aligned}\Delta \log C_{t+1} &= \mu_n \text{ w/prob } \pi_t, \\ &= \mu_d < \mu_n \text{ w/prob } 1 - \pi_t,\end{aligned}$$

where π_t is an iid variable determined at time t , and π_t and the realization of $\Delta \log C_{t+k}$ are statistically independent for all k . You can prove then that, if risk aversion > 1 , the P/D ratio will increase if consumption growth is equal to μ_d if and only if $\alpha > \theta$ i.e. the IES is small enough.

7 Savings and Portfolio Choice with Recursive Utility

Weil (QJE 1990, REStud 1993) shows how to solve portfolio problems with recursive utility. (Basically you can do guess and verify.) This allows to generalize naturally the results with CRRA utility, except now we see the two coefficients (RA and IES) showing up in different places. Weil studies for instance the effect of interest-rate uncertainty on savings. The Bellman equation is:

$$\begin{aligned}V(W) &= \max_c \left((1 - \beta)c^{1-\rho} + \beta (EV(W')^{1-\alpha})^{\frac{1-\rho}{1-\alpha}} \right)^{\frac{1}{1-\rho}}, \\ W' &= R(W - c), \\ \log R &\text{ iid } N(\mu, \sigma^2).\end{aligned}$$

In the case of $\rho \rightarrow 1$ we can write instead

$$V(W) = (1 - \beta) \log c + \frac{\beta}{\alpha} \log E \exp(\alpha V(W')),$$

which is the case that Weil solves analytically.

8 Hansen and Sargent (1980) Prediction Formulas

This is a useful formula which gives the expectation of a present value of a stochastic process, or revisions to this present value. Suppose we want to compute $E_t \sum_{j \geq 0} \beta^j x_{t+j}$. Write the Wold decomposition as $x_t = A(L)w_t$, where $A(L)$ is a lag operator. (For instance, $A(L) = 1 - \theta L$ and $x_t = w_t - \theta w_{t-1}$, or $A(L) = \frac{1}{1-\rho L}$ and $x_t - \rho x_{t-1} = w_t$.) Define the $[\cdot]_+$ operator as deletion of terms with negative powers, e.g. $[2L + 3 + 4L^{-1} - 3L^{-2}]_+ = 2L + 3$. Note that for any lag operator $B(L)$, we have $E_t(B(L)w_t) = [B(L)]_+ w_t$.

$$\begin{aligned} E_t \sum_{j \geq 0} \beta^j x_{t+j} &= E_t \sum_{j \geq 0} \beta^j A(L) \varepsilon_{t+j} \\ &= E_t \sum_{j \geq 0} \beta^j A(L) L^{-j} \varepsilon_t \\ &= \left[\sum_{j \geq 0} \beta^j A(L) L^{-j} \right]_+ \varepsilon_t \\ &= \left[\frac{LA(L)}{L - \beta} \right]_+ \varepsilon_t \\ &= \frac{LA(L) - \beta A(\beta)}{L - \beta} \varepsilon_t. \end{aligned}$$

The line before last computes the geometric series. The last line is more subtle: note that $\frac{LA(L) - \beta A(\beta)}{L - \beta} = D(L)$ is only a polynomial (i.e. positive lags), since we can factor a $L - \beta$ in the numerator. Since $\left[\frac{LA(L) - \beta A(\beta)}{L - \beta} \right]_+ = \left[\frac{LA(L)}{L - \beta} \right]_+ - \beta \left[\frac{1}{L - \beta} \right]_+$ and $\left[\frac{1}{L - \beta} \right]_+ = 0$, we have $\left[\frac{LA(L) - \beta A(\beta)}{L - \beta} \right]_+ = \frac{LA(L) - \beta A(\beta)}{L - \beta}$. (This is because $\frac{1}{L - \beta} = -\frac{1}{\beta} \frac{1}{1 - \frac{L}{\beta}}$ and since $1/\beta > 1$, this equation must be solved forward: $\frac{1}{1 - \frac{L}{\beta}} = \frac{L^{-1}}{L^{-1} - \frac{1}{\beta}} = \frac{L^{-1}\beta}{L^{-1}\beta - 1} = \frac{-L^{-1}\beta}{1 - L^{-1}\beta} = -L^{-1}\beta \sum_{k \geq 0} \beta^k L^{-k}$.)

Apply this formula to compute the expectation as of $t - 1$:

$$\begin{aligned} E_{t-1} \sum_{j \geq 0} \beta^j A(L) \varepsilon_{t+j} &= E_{t-1} E_t \sum_{j \geq 0} \beta^j A(L) \varepsilon_{t+j} = E_{t-1} \frac{LA(L) - \beta A(\beta)}{L - \beta} \varepsilon_t \\ &= \frac{\frac{LA(L) - \beta A(\beta)}{L - \beta} - \frac{-\beta A(\beta)}{-\beta}}{L} \varepsilon_{t-1} = \frac{LA(L) - \beta A(\beta) - A(\beta)(L - \beta)}{L(L - \beta)} \varepsilon_{t-1} \\ &= \frac{A(L) - A(\beta)}{L - \beta} \varepsilon_{t-1}. \end{aligned}$$

Thus we obtain a formula for the revision of expectations:

$$\begin{aligned} (E_t - E_{t-1}) \sum_{j \geq 0} \beta^j A(L) \varepsilon_{t+j} &= \left(\frac{LA(L) - \beta A(\beta)}{L - \beta} - \left(\frac{A(L) - A(\beta)}{L - \beta} \right) L \right) \varepsilon_t, \\ &= A(\beta) \varepsilon_t. \end{aligned}$$

This formula is much used, eg in the PIH-rational expectation literature (eg Quah (1989 JPE)).

Note how in the case of Campbell-Shiller, $A(L) = (I - AL)^{-1}$ and $A(\beta) = (I - A\beta)^{-1}$.